MULTIPLICATIVITY OF THE CHERN CHARACTER, WITH THE PRECISE BINOMIAL FORMULA FOR THE POWER SUMS OF THE CHERN ROOTS OF A TENSOR PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The splitting principle, which makes the Chern roots of a finite locally free module available. Same chapter chow.tex, Section \ref{section-splitting-principle}, invoked at proof line 9006. It reduces the computation to the situation where the Chern roots $x_{1,i}$ of $\mathcal{E}_1$ and $x_{2,j}$ of $\mathcal{E}_2$ can be named.
- The polynomials P_p used to define the Chern character. Same chapter chow.tex, Example \ref{example-power-sum}, cited at statement line 9001 and proof line 9014. Their shape, evaluated at the sums $x_{1,i} + x_{2,j}$, turns the binomial expansion into the stated identity in $A^p(X)$.

Source reference. Original reference 0F9B: sources/stacks-chow/pinned-original-source.tex, source lines 8673–8673. Navigation pointer for \ref{example-power-sum} used inside the counted span (line 9001); the argument that consumes it is retained in the span.

Source reference. Original section 02UK: sources/stacks-chow/pinned-original-source.tex, source lines 8422–8422. Navigation pointer for \ref{section-splitting-principle} used inside the counted span (line 9006); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup}

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used inside the counted span (line 8987); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $\mathcal{E}_1$ and $\mathcal{E}_2$ be finite locally free $\mathcal{O}_X$-modules. Then we have the equality*

$$ch(\mathcal{E}_1 \otimes_{\mathcal{O}_X} \mathcal{E}_2) = ch(\mathcal{E}_1)ch(\mathcal{E}_2)$$

More precisely, we have

$$P_p(\mathcal{E}_1 \otimes_{\mathcal{O}_X} \mathcal{E}_2) = \sum\nolimits_{p_1+p_2=p} \binom{p}{p_1} P_{p_1}(\mathcal{E}_1)P_{p_2}(\mathcal{E}_2)$$

in $A^p(X)$ where P_p is as in Example 0F9B.

Proof. It suffices to prove the more precise statement. By Section 02UK this follows because if $x_{1,i}$, $i = 1, \ldots, r_1$ and $x_{2,i}$, $i = 1, \ldots, r_2$ are the Chern roots of $\mathcal{E}_1$ and $\mathcal{E}_2$, then $x_{1,i} + x_{2,j}$, $1 \leq i \leq r_1$, $1 \leq j \leq r_2$ are the Chern roots of $\mathcal{E}_1 \otimes \mathcal{E}_2$. Hence we get the result from the binomial formula for $(x_{1,i} + x_{2,j})^p$ and the shape of our polynomials P_p in Example 0F9B. □